

# Curriculum Vitæ

Jules Berry – Postdoctoral Researcher

## Personal Information

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Jules Berry.

Postdoctoral Researcher in Applied Mathematics at Université Paris-Saclay – CentraleSupélec, France.

**Languages:** French (native), English (fluent)

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## Academic Positions

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Since 11/2025     **Postdoctoral Researcher**, L2S – Université Paris-Saclay – CentraleSupélec, France.  
**Supervisors:** Riccardo Bonalli (L2S – CNRS) and Laurent Pfeiffer (L2S – INRIA).  
**Research topic:** Risk-averse stochastic optimal control.

## Education

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2022 – 2025     **Ph.D. in Mathematics and their Interactions**, IRMAR – Université de Rennes – INSA Rennes, France.  
**Thesis:** Contributions to the study of second order mean field games: approximation of stable solutions and sticky processes on networks – [tel-05393129](tel:05393129).  
**Defense date:** October 23, 2025 at INSA Rennes.  
**Advisors:** Olivier Ley (IRMAR – INSA Rennes) and Francisco Silva (XLIM – Université de Limoges).  
2020–2022     **Master in Fundamental Mathematics**, Université de Rennes, France, with Highest Honors.

## Publications

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- [J4] Berry, J., Ley, O., & Silva, F. J. (2026). [A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem via metric regularity techniques and applications to nonlinear PDEs](#). *Foundations of Computational Mathematics*.
- [J3] Berry, J., & Colantoni, F. (2026). [Sticky diffusions on star graphs: characterization and Itô formula](#). *Stochastic Processes and their Applications*.
- [J2] Berry, J., & Camilli, F. (2025). [Stationary Mean Field Games on networks with sticky transition conditions](#). *ESAIM: Control, Optimisation and Calculus of Variations*.
- [J1] Berry, J., Ley, O., & Silva, F. J. (2025). [Approximation and perturbations of stable solutions to a stationary mean field game system](#). *Journal de Mathématiques Pures et Appliquées*.

## Preprints

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- [P2] Basti, A., Berry, J. and Camilli, F. (2026). [Markov Chain Approximation of Sticky Diffusions and Hamilton-Jacobi-Bellman Equations on Networks](#), hal-05706139.
- [P1] Berry, J. (2025). [Some error estimates for semidiscrete finite element approximations of stable solutions to mean field game systems](#), hal-05365007.

## Communications

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### Invited talks

- Sept. 2025 **Durham Symposium on Mean Field Games**, *Approximation of stable solutions to second order mean field game systems*, Durham, United Kingdom.
- Oct. 2024 **ANR COSS Workshop**, *Sticky diffusion processes on networks and corresponding Mean Field Games*, Rennes, France.
- Feb. 2024 **Rennes-Tours Workshop**, *Approximation of stable solutions to a stationary mean field game system*, Rennes, France.

### Talks in seminars

- Feb. 2026 **Séminaire EDP**, *Mean field games of sticky diffusions on networks*, Laboratoire de mathématiques de Besançon, Besançon, France.
- Dec. 2024 **MOD Seminar**, *Sticky diffusion processes on networks and corresponding Mean Field Games*, XLIM, Limoges, France.
- April 2024 **Seminario di Modellistica Differenziale Numerica**, *Approximation of stable solutions to second order mean field game systems*, La Sapienza Università di Roma, Rome, Italy.

### Contributed talks

- Jun. 2026 **Congrès National d'Analyse Numérique (CANUM)**, *A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem*, Saint-Jacut-de-la-mer, France.
- Mar. 2026 **Journées SMAI MODE**, *Approximation of stable solutions to second order mean field game systems*, Nice, France.
- Mar. 2026 **Congrès des Jeunes Chercheurs en Mathématiques Appliquées**, *A nonsmooth extension of the Brezzi-Rappaz-Raviart approximation theorem*, Champs-sur-Marne, France.
- Sep. 2025 **European Conference on Numerical Mathematics and Advanced Applications (ENUMATH)**, *Approximation of stable solutions to second order mean field game systems*, Heidelberg, Germany.
- Jan. 2024 **3rd International Conference on Variational Analysis and Optimization – In Honor of Boris Mordukhovich**, *A theorem of Brezzi, Rappaz, and Raviart from the point of view of variational analysis*, Santiago, Chile.
- Jan. 2024 **Conference on Numerical methods for optimal transport problems, mean field games, and multi-agent dynamics**, *Approximation of stable solutions to a stationary MFG system*, Valparaíso, Chile.

### Posters

- Nov. 2025 **Première rencontre nationale du RT Optimisation**, *Mean Field Games on networks with sticky transition conditions*, Lyon, France.
- Juin 2024 **Summer School on Machine Learning and Optimal Control**, *Approximation of non-differentiable nonlinear problems*, Gaeta, Italie.
- Mars 2024 **Journées SMAI MODE**, *Approximation of stable solutions to a stationary mean field game system*, Lyon, France.

## Teaching

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- 2026 **Introduction to stochastic calculus**, ENSTA Paris-Saclay, Tutorial sessions (10h).  
Topics: Brownian motion, stochastic integrals, stochastic differential equations, Girsanov's theorem.
- 2022–2025 **Analysis 3**, INSA Rennes, Tutorial sessions (48h/year), Undergraduate level (2nd year).  
Topics: improper integrals, numerical series, power series, Fourier series, differential calculus.
- Analysis Tools for Engineering**, INSA Rennes, Tutorial sessions (10h/year),  
Undergraduate level (3rd year).  
Topics: Lebesgue integrals, Fourier transform, complex analysis.

## Academic Supervision

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### Master's Students

- 11/2025 – 4/2026 **Chaimae El Omari**, research initiation project, CentraleSupélec.  
Co-supervised with Laurent Pfeiffer (L2S – INRIA).  
Topic: Potential mean field games with non-differentiable Hamiltonians.
- 11/2024 – 12/2024 **Angelina Jammart**, Master's student seminar, Université de Rennes.  
Co-supervised with Othmane Jerhaoui (IRMAR – INSA Rennes).  
Topic: Introduction to viscosity solutions for Hamilton-Jacobi equations.

## Funding

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- 2025 **SMAI BOUM**, Funding for a week of collaborative research at CIRM.  
Project: Comparison of macroscopic models for crowd motion.  
In collaboration with Théo Girard (Univ. Tours) and Florian Peru (Univ. Franche-Comté).

## Awards

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- 2024 **Poster Award**: First prize ex-aequo, SMAI MODE Conference, Lyon.

## Research Visits

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- Jul. 2026 **Université Marie-et-Louis-Pasteur**, Besançon, invited by Carlotta Donadello and Nathaël Alibaud, 3 days.
- March–May 2024 **Università di Roma la Sapienza**, invited by Fabio Camilli, 3 months.
- 2022–2025 **Université de Limoges**, invited by Francisco Silva, 6 one-week visits.

## Institutional Activities

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- 2024–Present **Reviewer** for *J. Math. Anal. Appl.* (1), *NoDEA* (1), *Electron. J. Probab.* (1).